POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2015
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

	a)	Write the advantages of digital signals over analog signals. Define Energy and Power signals.	8	1
	b)	Show that the impulse response of a BIBO stable LTI system is absolutely summable.	7	2
2.	a)	Find the convolution between two signals $x[n] = u[n]$ and $h[n] = a^n u[n]$, $0 < a < 1$	8	2
	b)	Find the Z-transform of $x[n] = a^n \cos w_0 n\, u[n]$	7	3
3.	a)	Find the 8-point DITFFT of $x(n) = \sin\frac{3\pi n}{4}$, $for\ 0 \le n \le 7$	8	4
	b)	What is Zero Padding ? Find the circular Convolution of $x_1[n] = \{1,2\}$ & $x_2[n] = \{3,2,1\}$	7	4
4.	a)	Obtain the parallel form realization of following IIR filter	7	5

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

	b)	Convert the following IIR filter into lattice ladder structure	8	5

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

5.	a)	Design a lowpass filter which will have -3 dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec.	8	
	b)	Design a linear FIR filter using Kaiser window to meet the following specifications:	7	6/3

$$0.99 \le |H(e^{j\omega})| \le 1.01,\ for\ 0 \le |\omega| \le 0.19\pi$$
$$\le |H(e^{j\omega})| \le 0.01,\ for\ 0.21\pi \le |\omega| \le \pi$$

6.	a)	Determine H(z) using impulse invariance method for the following analog	7	7.2

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filter's transfer function :

$$H_a(s) = \frac{1}{(s + 0.5) + (s^2 + 0.5s + 2)}$$

b) Design a second order discrete-time Butterworth filter with cut-off frequency of 1 KHz and sampling frequency of 10^4 samples/sec by using bilinear transformations. Also plot the poles of the filter.

7. Write short notes on: (Any two)

a) Region of Convergence

b) Causal and Non-Causal system

c) Gibbs Phenomena

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